

Structural bottlenecks impairing neurological recovery

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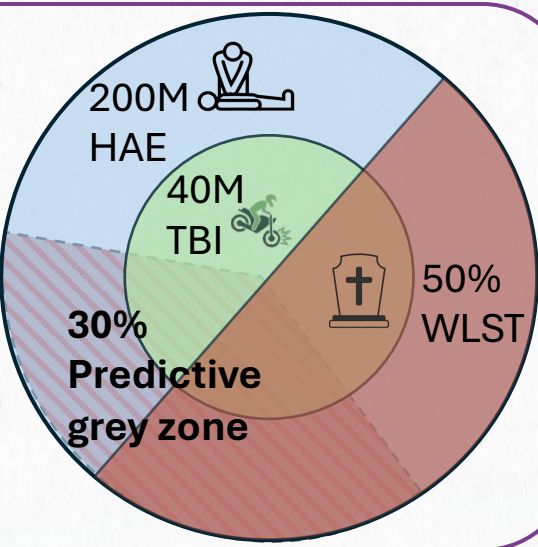
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Context

DALY = Disability-Adjusted Life Years

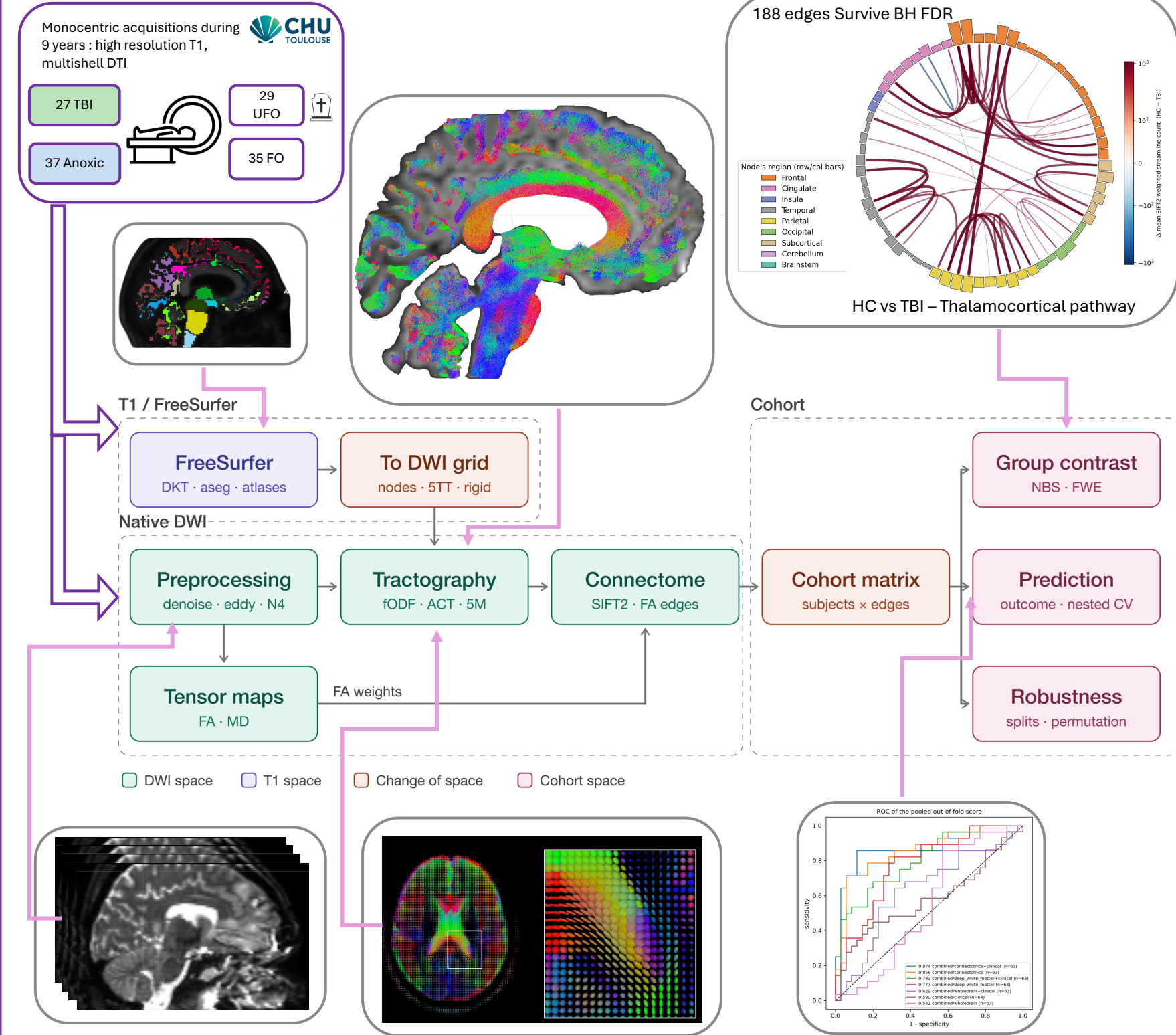
By year in European Union



Added value

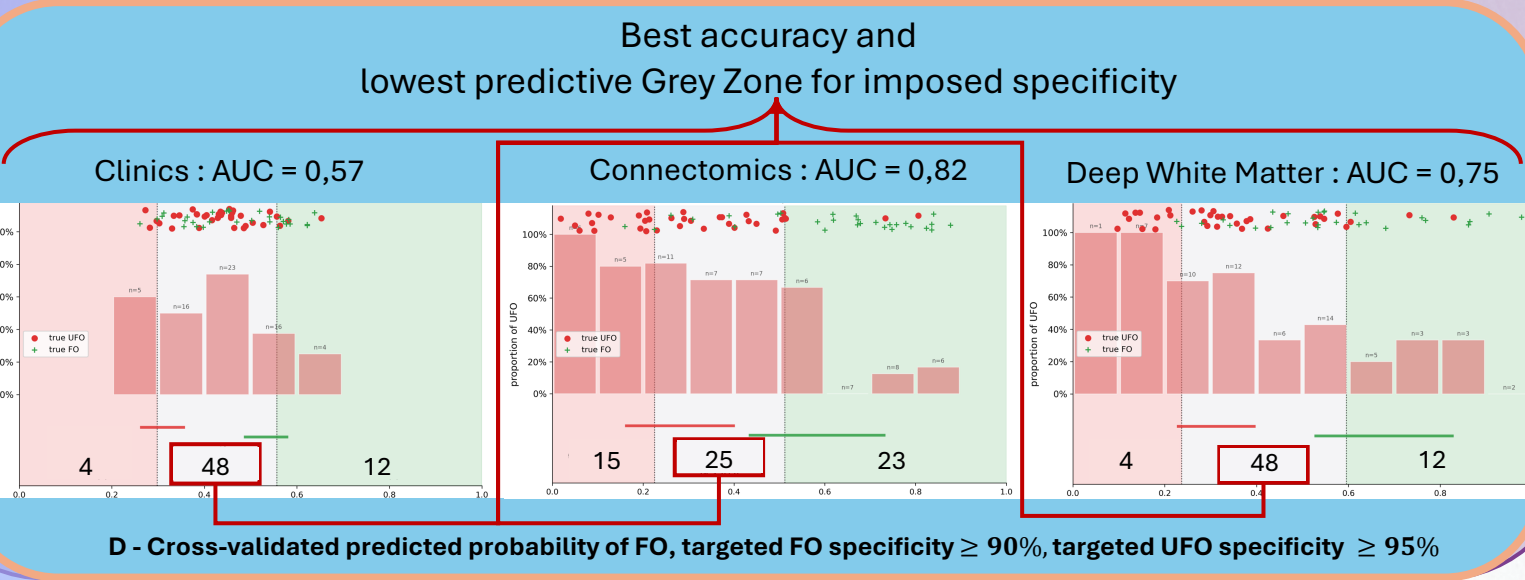
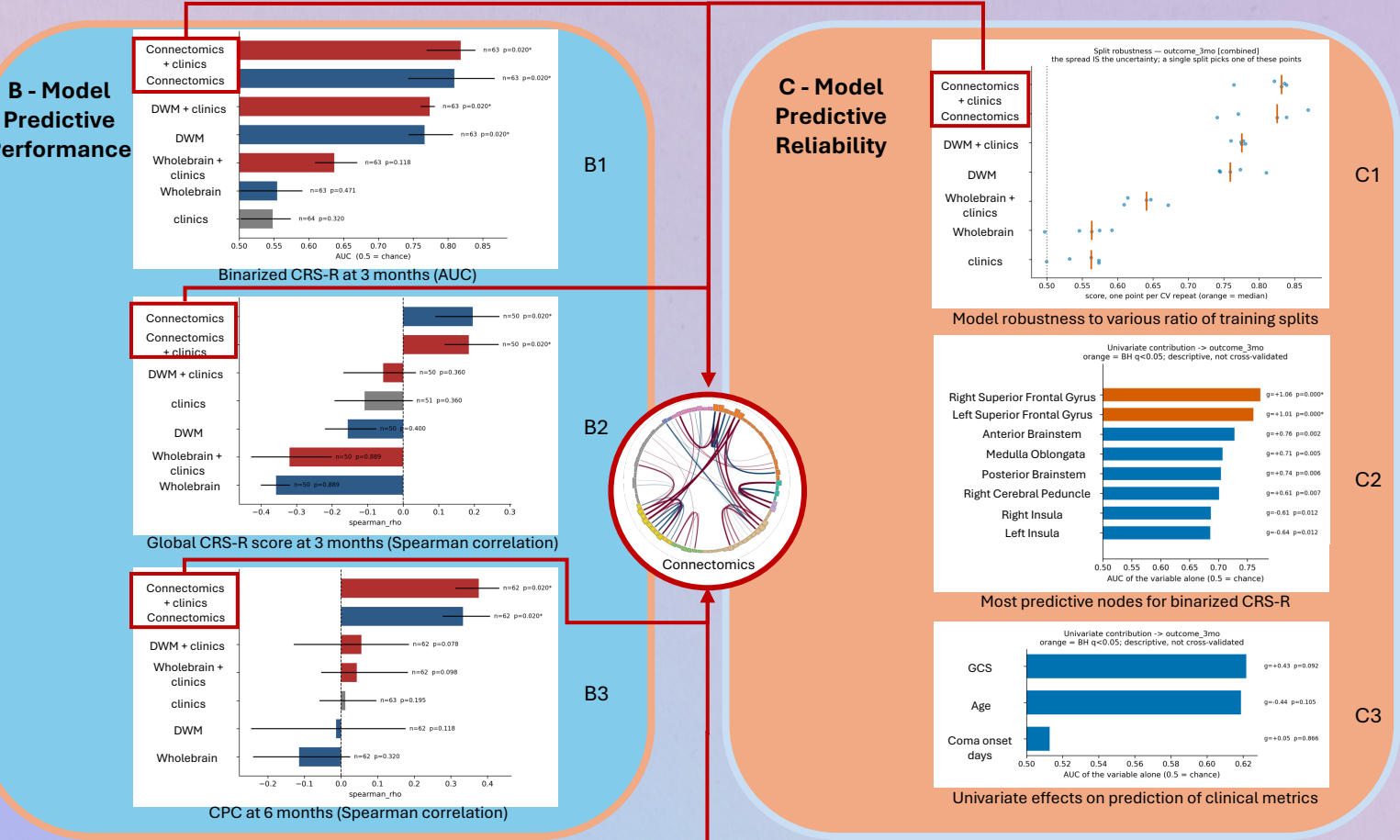
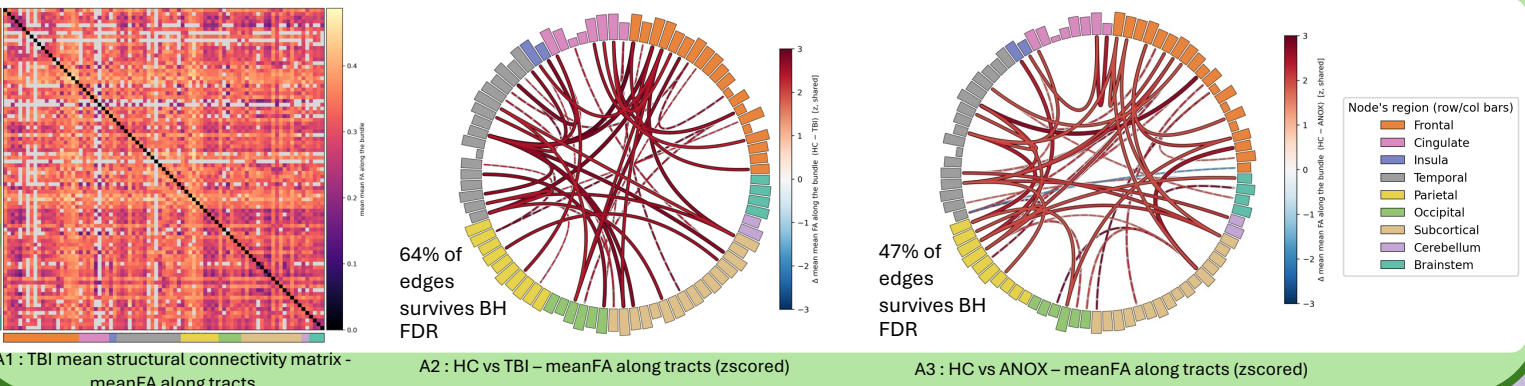
- Connectomics halves predictive grey zone
- Structural bottlenecks are measurable at 3T in clinical settings
- Explainable by design

Methods



Results

A – Matrices and chord plots – Connectivity at a glance



Conclusion

- Topology outperform regional average
- Grey zone divided by two
- AUC Connectomics > AUC Deep White Matter > AUC Clinics
- One pipeline for two aetiologies
- Anatomically interpretable best candidate networks

References

Silva 2017, Critical Care Medecine ; Silva 2015, American Academy of Neurology ; Puybasset 2022 Intensive Care Medecine ; Velly 2018 The Lancet Neurology ; Edlow 2024 Science Translational Medecine ; GBD Collaborators 2021 The Lancet

